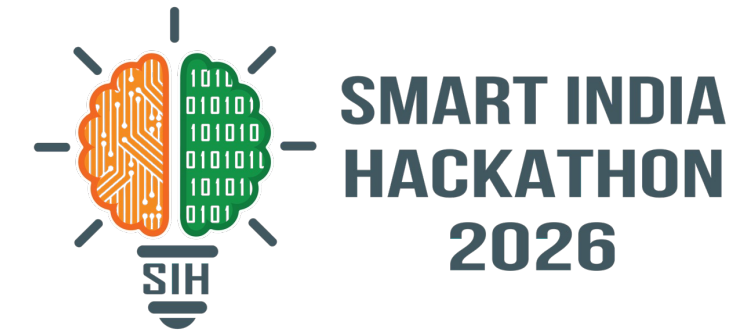


SMART INDIA HACKATHON 2026



GEOVAX — Intelligent Harmonisation of Multi-source Geospatial Data

- **Problem Statement ID** – SIH26013
- **Problem Statement Title** – Automated Integration and Intelligent Harmonisation of Multi-source Geospatial Data for Urban Land Record Management
- **Theme** – Smart Automation
- **PS Category** – Software
- **Team ID** – SIH1167
- **Team Name** – Perpetually Curious



Four to ten authoritative geometries per parcel — and they disagree.

THE CHALLENGE

Four to ten geometries per parcel

Reconciled by hand in GIS

No record of why a decision was made

Errors propagate silently

Existing gap in current GIS workflows

Slow cadastral finalisation

Data siloed per department

Averaging invents boundaries

OUR SOLUTION

An AI-enabled platform that adjudicates — and shows its working

Claims, Not Records

Dempster–Shafer Evidence Fusion

Never Invents a Boundary

ULPIN + Provenance Ledger

WHY IT MATTERS

AI/ML Spatial Matching

Automated Topology Correction

Intelligent Attribute Mapping

Georeferencing & CRS

Change Detection & Typing

Confidence Scoring, Graded A–E

107,262
Real features harmonised end to end

1.51 m @ 073°
Systematic inter-departmental offset recovered

99.988%
Candidate pairs eliminated by blocking

43.4%
Conflicts auto-resolved, the rest queued

14,702
ULPINs minted, every one unique

17 min 45 s
One full run on 2 vCPU / 7 GB

0
Synthetic features anywhere in the corpus

TECHNICAL NOVELTY: EVIDENCE-BASED HARMONISATION CORE

✓ Claims are never deleted — every decision stays re-examinable

✓ Mass sits on “unknown”, so ignorance is visible, not hidden

✓ Selects an observed boundary, never an averaged one

✓ Bounded, audited topology repair — refuses beyond 0.5 m

✓ Six-dimension explainable confidence, graded A–E per feature

BUSINESS NOVELTY: A DEFENSIBLE RECORD, NOT A FASTER GUESS

✓ ULPIN (Bhu-Aadhaar) minting with split / merge genealogy

✓ Merkle-anchored ledger a citizen can verify independently

✓ Adjudication queue batched by cause, ranked by decision value

✓ Runs offline on 2 vCPU — no data centre, no external service

✓ OGC API - Features: works in QGIS with no custom code

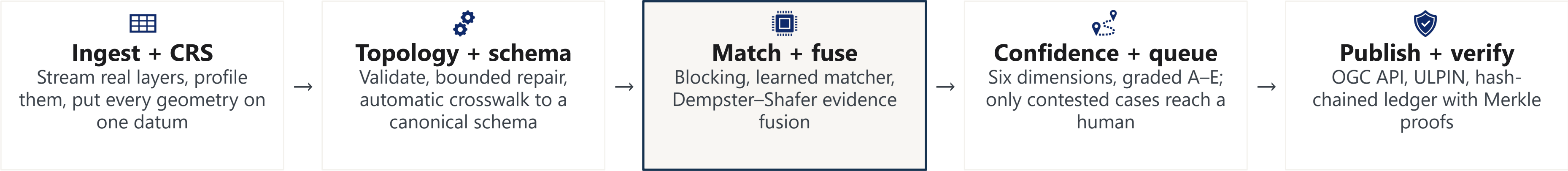
PROBLEM WE ADDRESS

Harmonisation of multi-source land data is manual, slow, error-prone — and preserves no reason for any decision.



We automate that adjudication so every published boundary carries **its evidence, its confidence and its provenance**.

From Disagreeing Sources to a Defensible Land Record



 CORE GEOSPATIAL ENGINE**Coordinate & Datum Engine**

- Everest 1830 / Kalianpur family
- EPSG:7755 India NSF LCC
- GCP ladder + Baarda test

Topology

- 15 validation rules
- Bounded repair, refused beyond 0.5 m
- Planar-partition gap scan

Spatial Matching

- Grid + STRtree blocking
- Median offset removal

Evidence Fusion

- Dempster–Shafer mass construction
- Independence discounting
- Statutory rules R-GEO-01 / R-POR-01
- Conflict mass retained

Confidence & Identity

- Six dimensions, graded A–E
- ULPIN checksum + genealogy
- Hash chain, Merkle inclusion proofs



PLATFORM ARCHITECTURE

PRODUCING AUTHORITIES

Drone / NAKSHA · Revenue · Municipal GIS · Utilities · GNSS · GeoAI

1 · INGEST AND PROFILE

Format connectors | Dataset profiler | CRS + datum engine | GCP rubber-sheeting
Claims, not records — nothing entering the platform is treated as fact

2 · HARMONISATION CORE

- ✓ Topology validate + repair
- ✓ Automatic schema matching
- ✓ Indic transliteration
- ✓ AI/ML spatial matching
- ✓ Global assignment + cardinality
- ✓ Entity clustering

3 · DECISION LAYER

Which claim survives, and why — recorded either way

A. EVIDENCE FUSION

- Dempster–Shafer combination
- Independence discounting
- Geometric fusion by evidence
- 629,660 claims fused

Decides what survives

B. STATUTORY RULE ENGINE

- R-GEO-01 — > 10 m refused
- R-POR-01 — poramboke findings
- R-CTL-01 — control precedence
- 702 refusals in this run

Encodes the law

C. ADJUDICATION QUEUE

- 20,207 cases escalated
- Batched by cause, not by case
- Ranked by decision value
- Every decision becomes a label

Where humans decide

CONFIDENCE SCORING

Six dimensions, explainable, graded A–E

4 · PUBLICATION

- PostGIS canonical store
- Provenance ledger + Merkle root
- OGC API - Features / STAC
- Subscription bus · Web-GIS console



TECHNOLOGY STACK

Python 3.11, Apache-2.0

12,700 lines · 30 modules · 81 passing tests

PostgreSQL + PostGIS

canonical store, exact topological predicates on 6 M parcels

Shapely / GEOS + STRtree

validation, bounded repair, blocking, spatial joins

PROJ / pyproj

Indian datum family, EPSG:7755, EPSG:32644 measurement CRS

scikit-learn HistGradientBoosting

weak-supervised, self-trained spatial matcher

rasterio / GDAL COG · CityJSON 1.1

raster tier, terrain chain, LOD1 city model

FastAPI · OGC API - Features · MapLibre

platform API and the operator console



OUTPUT SURFACES

OGC OGC API - Features + GeoJSON

MVT Vector tiles · Web-GIS console

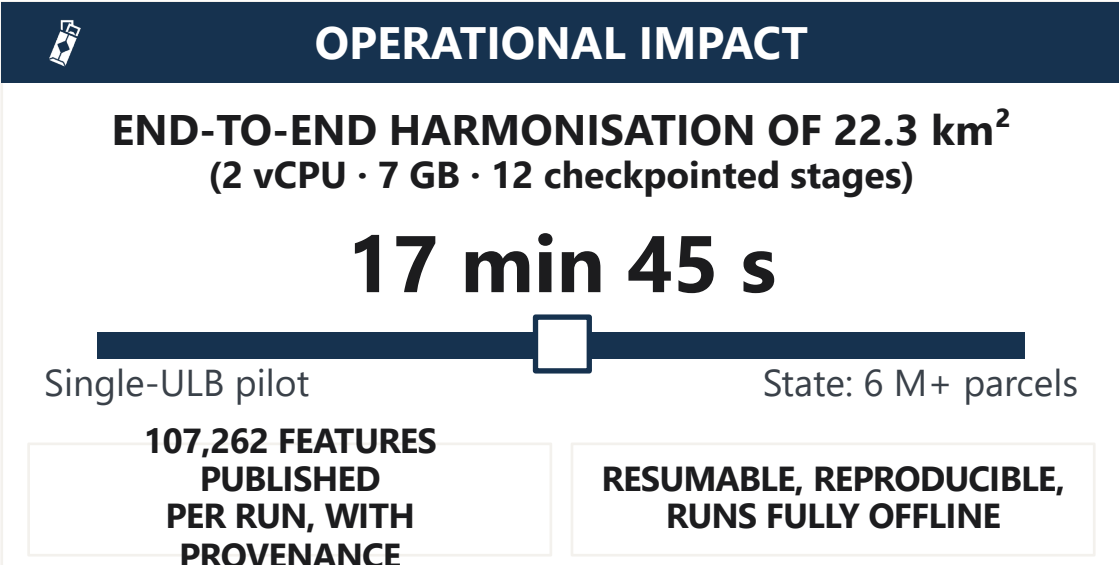
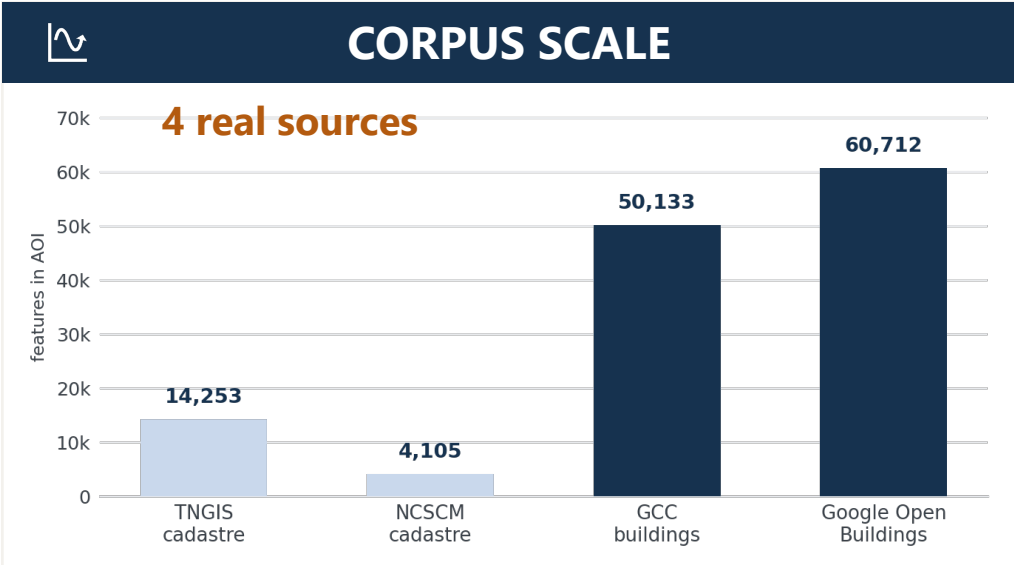
ULP ULPIN register with genealogy

LOD1 CityJSON 1.1 3-D city model

QUE Adjudication queue + GeoJSON

LGR Provenance ledger + Merkle root

IMPACT & BENEFITS



ECOSYSTEM & ADOPTION			
PROGRAMMES	DILRMP	NAKSHA	AMRUT / Bhuvan
REGISTRIES & POLICY	LGD Directory	Nat. Geospatial Policy 2022	DPDP Act 2023
OPEN STANDARDS	OGC API - Features · GeoJSON · CityJSON 1.1 · ISO 19107 / 19115		
DEPARTMENTS SERVED	Revenue / Survey	Municipal ULB	DoLR / State GIS

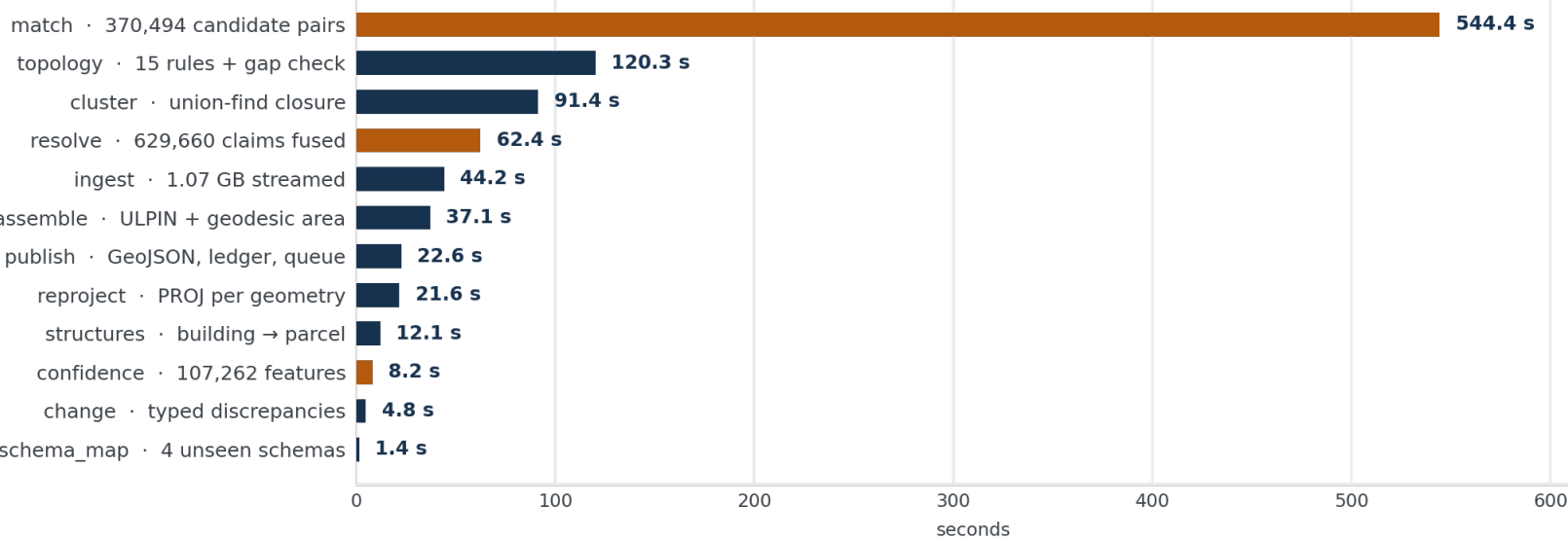
22.3 km ² Central Chennai harmonised per run	6.02 M TNGIS parcels ingested statewide	43.4% Conflicts closed with no human involved	0.9997 Mean topological confidence across the AOI	100% ULPIN single-character errors detected	0 Boundaries invented by averaging
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TECHNICAL BENEFITS	OPERATIONAL BENEFITS	STRATEGIC BENEFITS	STANDARDS AND STATUTORY CONFORMANCE			
✓ Reproducible, checkpointed runs ✓ Refusals recorded, never silent ✓ Policy replayed without re-ingest ✓ Runs offline on district hardware	✓ Manual reconciliation removed ✓ Queue batched by cause, not case ✓ Survey effort aimed by evidence ✓ Officer decisions become training data	✓ NAKSHA readiness measured per ward ✓ Exchange by open standard, not export ✓ Citizen-verifiable record integrity ✓ Sovereign, on-premises, no lock-in	Instrument	Type	Status	How GEOVAX conforms
			DILRMP	Programme	✓ Built	14-character ULPIN with checksum; split / merge genealogy carries encumbrance across mutation
			NAKSHA — 0.5 m urban spec	Programme	✓ Measured	Positional confidence scored against the spec; the shortfall is reported per ward, not hidden
			LGD hierarchy (MoPR)	Registry	✓ Built	state / district / taluk / village LGD codes are the canonical join key
			Nat. Geospatial Policy 2022	Policy	✓ Built	EPSG:7755 India NSF LCC is a first-class target CRS; open standards throughout
			DPDP Act 2023	Statute	✓ Built	Owner attributes absent from the feature API; purpose-bound, logged access only
			OGC API - Features Part 1	Standard	✓ Conformant	/conformance, /collections, /items — QGIS works with no custom code
			CityGML LOD1 / CityJSON 1.1	Standard	✓ Built	Solid geometry, measuredHeight, storeysAboveGround — 61,066 extruded
			STAC catalogue	Standard	⚙️ Designed	Item structure designed, not built here — stated rather than implied

WHERE IT APPLIES	Cadastral finalisation	Mutation & subdivision	NAKSHA survey planning	Encroachment findings	Development control	Utility coordination	Inter-dept. exchange	Citizen verification
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ONE 17-MINUTE RUN · 107,262 FEATURES · 2 vCPU
AND THE 56.6% IT REFUSES TO CLOSE WITHOUT A HUMAN

Wall-clock by pipeline stage — 1,065 s (17 min 45 s) end to end on 2 vCPU / 7 GB



12 checkpointed stages on 2 vCPU / 7 GB. Matching dominates because 370,494 candidate pairs are scored — after blocking removed 99.988% of the 3.04×10^9 possible.



VIABILITY

- ✓ **Licensing** — Apache-2.0, and every input layer is CC0-1.0 or CC-BY-4.0 — nothing blocks a deployment.
- ✓ **Deployment** — Python 3.11 + PostGIS via docker-compose; runs offline in a district office on 2 vCPU.
- ✓ **Adoption path** — OGC API - Features, so a department works against it in QGIS with no GEOVAX-specific code.
- ✓ **Sizing** — A measured path from a single-ULB pilot (< 500 k parcels) to Tamil Nadu statewide (6 M+).
- ✓ **Maintainability** — 81 passing tests, per-stage checkpoints, an AOI manifest with SHA-256 for every input.
- ✓ **Institutional fit** — The LGD hierarchy is the join key throughout, so multi-ULB isolation is configuration, not a rewrite.



ACCURACY

- Matcher AUC 1.0000, expected calibration error 0.0002 over 5-fold CV
- 1.51 m @ 073° offset recovered from 5,226 pairs, residual RMS 2.51 m
- Mean topological confidence 0.9997; ULPIN checksum catches 100% of typos
- nDSM median 0.023 m — the derived ground surface sits on the ground



EFFICIENCY

- Blocking removes 99.988% of 3.04×10^9 pairs before a feature is computed
- A 1.07 GB cadastral file is streamed in ~40 MB of RAM, not loaded
- 107,262 features harmonised in 1,065 s on 2 vCPU — no GPU, no cluster



PERFORMANCE

- 43.4% of 35,690 conflicts auto-resolved; 20,207 escalated, batched into 2 causes
- Change detector's measured false-positive rate 1.79% on a null-change experiment
- 61,066 buildings extruded to LOD1; 31,494 with no height emitted flat and flagged



RISKS AND MITIGATION

- **No independent ground truth** — CV figures are against the platform's own weak labels. A pilot's first week is a 200–300 point control survey; everything downstream then becomes empirical.
- **Declared accuracies may be aspirational** — Every fusion weight is anchored to them — which is why ground truth comes first.
- **A queue nobody works** — 56.6% needs a human. Batching by cause makes the workload tractable; it cannot make it zero.
- **No Indian float DSM openly available** — The terrain chain is proved on a real calibrated surface (Geobasis NRW DOM1) and labelled as such.
- **AMRUT layer contributes zero features here** — Reported as a completeness gap, not silently run with three sources where four were configured.

Read the middle track first — what is designed but not built is stated, not implied



BUILT AND MEASURED

STAGE 1 ✓ built

Ingest + CRS



STAGE 2 ✓ built

Topology + schema



STAGE 3 ✓ built

Match + fuse



STAGE 4 ✓ built

Confidence + queue



STAGE 5 ✓ built

Publish + ledger



DESIGNED, NOT BUILT

ITEM A

STAC catalogue



ITEM B

Subscription bus



ITEM C

Adjudication write path



DEPLOYMENT PATH

NEXT 1

Ground-truth control survey



DATA SOURCES AND STANDARDS

- **TNGIS Tamil Nadu cadastrals** — TNeGA · CC0-1.0 · 6.02 M parcels · github.com/ramSeraph/indian_cadastrals
- **NCSCM coastal cadastrals** — MoEFCC · CC0-1.0 · 220,668 parcels · ncscm.res.in
- **Greater Chennai Corporation footprints** — GCC via TNGIS · CC0-1.0 · 964,053 · github.com/ramSeraph/indian_buildings
- **Google Open Buildings v3** — CC-BY-4.0 / ODbL-1.0 · sites.research.google/gr/open-buildings
- **AMRUT / Bhuvan urban buildings** — NRSC, ISRO · bhuvan.nrsc.gov.in
- **Geobasis NRW DOM1 float DSM** — Bezirksregierung Köln · dl-de/by-2-0 · 1 m grid · opengeodata.nrw.de
- **Local Government Directory** — Ministry of Panchayati Raj · lgdirectory.gov.in
- **OGC API - Features Part 1: Core** — ISO 19168-1 · ogcapi.ogc.org/features
- **Formats and quality standards** — GeoJSON · CityJSON 1.1 · COG · ISO 19107 / 19115 / 19157 · rfc-editor.org/rfc/rfc7946
- **Statute and programme** — DILRMP · NAKSHA (DoLR) · NGP 2022 · DPDP Act 2023 · TN CDBR 2019 · Sol CORS

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